Generative AI-Based Cloud Removal and Reconstruction for LISS-IV Satellite Imagery

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Abstract— Cloud contamination reduces the availability of optical satellite imagery for land-cover mapping, agriculture, forest monitoring, and disaster assessment. This paper presents a research framework for generative-AI-based cloud removal and reconstruction tailored to Indian Remote Sensing Satellite Linear Imaging Self Scanner-IV (LISS-IV) imagery. The design combines cloud-mask estimation, multiscale spatial feature extraction, optional temporal and SAR guidance, and conditional generative reconstruction. Unlike purely visual enhancement, the framework preserves observed clear pixels and evaluates spectral consistency, structural similarity, and uncertainty. A comparative study of 30 relevant research and foundational sources identifies trade-offs among interpolation, temporal gap filling, U-Net/GAN methods, SAR-optical fusion, transformers, and diffusion models. This is a methodology and literature synthesis; no experimental scores are claimed because a verified LISS-IV dataset and model outputs were not supplied. A reproducible evaluation protocol is defined using PSNR, SSIM, RMSE, spectral angle mapper, cloud-region metrics, latency, and downstream consistency. Particular attention is given to sensor-domain shift, limited paired data, geospatial alignment, and the risk that plausible generated textures may not represent the true land surface.

Index Terms— cloud removal, LISS-IV, remote sensing, generative AI, conditional GAN, diffusion model, SAR-optical fusion, image reconstruction.

I. INTRODUCTION

Optical Earth-observation imagery supports land-cover mapping, crop monitoring, environmental assessment, and emergency response. Cloud and cloud-shadow contamination creates missing or distorted observations; thick clouds can fully obscure surface information. Cloud removal is therefore not simply cosmetic enhancement: reconstructed pixels can influence scientific conclusions and operational decisions. Single-image methods infer missing content from spatial context, while multitemporal approaches exploit observations of the same region on other dates. SAR imagery offers complementary structural information under cloudy conditions, although radar backscatter is not a direct substitute for optical reflectance [4], [6], [16].

LISS-IV is a high-spatial-resolution multispectral sensor in India's remote-sensing ecosystem. A method developed on Sentinel-2 cannot be assumed to transfer directly to LISS-IV because band definitions, spatial resolution, radiometry, noise, and acquisition geometry differ. Sensor-specific adaptation is therefore a central requirement rather than a simple input-file change. The system should maintain georeferencing and original clear observations, and identify generated regions so downstream users can distinguish measured from inferred pixels.

The project objectives are to detect cloud-contaminated pixels; reconstruct hidden surface content; preserve clear areas and spectral relationships; export georeferenced outputs; and compare classical and generative methods using a fair protocol. The work has four contributions: a sensor-aware LISS-IV pipeline; a 30-source comparative synthesis; a multi-objective loss and evaluation plan; and a reproducibility checklist. This manuscript is a research framework, not a claim of completed experiments or an IEEE-published paper.

II. RELATED WORK

A. Classical and temporal reconstruction

Spatial interpolation methods such as nearest-neighbour, bilinear, bicubic, and inverse-distance weighting are inexpensive but tend to blur large missing regions and cannot recover structures absent from local context. Temporal gap filling is stronger when a clear observation exists near the target date. However, crop cycles, floods, construction, seasonal vegetation, and illumination differences can make older reference pixels inaccurate. Tensor-factorization methods exploit relationships across time and spectral dimensions and can be robust to imperfect masks [7]. Comparative gap-filling studies motivate evaluating spatial, temporal, spatiotemporal, and spatio-spectral strategies rather than selecting a deep model based on appearance alone [19].

B. CNN and U-Net approaches

U-Net encoder–decoder structures are useful for cloud-mask prediction and image-to-image reconstruction because skip connections preserve local detail [27]. CNNs are computationally practical and can serve as strong baselines. Their limitations include oversmoothing under pixel losses, limited ability to infer large hidden structures, and sensitivity to training data distribution. Scene-level train/test splitting is essential because neighbouring patches from the same acquisition are highly correlated.

C. GAN-based restoration

Generative adversarial networks train a generator to reconstruct clear imagery and a discriminator to distinguish generated samples from reference imagery [26]. Zheng et al. combine U-Net and GAN-based restoration in a single-image cloud-removal framework [1]. Transfer learning may reduce data requirements [12], and unpaired image translation provides a possible route when exact cloudy/clear pairs are unavailable [25]. However, adversarial losses can encourage plausible but false texture. A generated road, river, or field boundary may look realistic but still be geospatially wrong. GAN outputs must therefore be checked with spectral and structural metrics rather than visual sharpness alone.

D. SAR-optical and temporal fusion

SAR can provide structural cues beneath clouds because microwave imaging is not obstructed in the same way as optical imaging. Meraner et al. combine SAR-optical fusion with a deep residual network and a cloud-adaptive loss [16]. Multitemporal Sentinel-1 SAR and Sentinel-2 optical images can help account for ground changes [17], while SEN12MS-CR-TS provides multimodal and multitemporal resources [4], [20], [21]. Recent methods use cross-attention or hybrid 3-D/2-D feature extraction [8], [9]. The central challenge is modality mismatch: radar and optical sensors encode different physical properties. Fusion must be spatially aligned, normalized, and gated rather than concatenated blindly.

E. Diffusion and attention

Conditional diffusion models progressively denoise samples under conditioning information. DiffCR proposes conditional guidance and a time-condition fusion block [3]; diffusion enhancement has also been studied for ultra-resolution imagery [2], and a dual-branch residual design adds discrepancy correction [5]. Progressive diffusion and unsupervised latent disentanglement are further directions [10], [11], [15]. Attention can capture long-range relationships across bands and modalities [8], [29]. Their costs include compute, inference latency, and training-data demands. A compact U-Net baseline followed by a conditional GAN or diffusion experiment is a practical staged plan for a student team.

III. PROPOSED SYSTEM

The system has four stages: (1) data preparation and co-registration; (2) cloud and shadow mask estimation; (3) conditional reconstruction; and (4) preservation, uncertainty estimation, and export. Inputs are cloudy multispectral image X, cloud mask M, and optional auxiliary information A, such as a near-date clear optical observation or co-registered SAR image. Outputs are reconstructed image Y and confidence map U. The model should preserve observed clear pixels and modify only contaminated pixels wherever possible.

A. Data preparation

All scenes should be converted to a documented radiometric scale, checked for invalid pixels, and aligned to a common coordinate reference system. Images are tiled into fixed-size patches with overlap to reduce boundary artefacts. Training splits should be made by geographic scene or acquisition date, not by random neighbouring patches, to avoid leakage. Band order, spatial resolution, no-data values, projection, and scaling factors must be recorded. If exact paired cloudy/clear LISS-IV scenes are unavailable, synthetic clouds can be applied to clear images to create paired supervision. This is useful for initial benchmarking but cannot fully represent real cloud thickness, shadows, haze, and sensor artefacts. Real-cloud validation remains necessary.

B. Cloud-mask estimation

A U-Net-style mask predictor estimates cloud probability M■, including cloud shadows where labels are available. Mask quality is critical: an overly conservative mask leaves cloud remnants, while an overly broad mask alters clear ground. The model should distinguish thin cloud, thick cloud, shadow, snow/bright roofs, and no-data regions where training labels permit. Evaluation should report precision, recall, and intersection-over-union for the mask separately from reconstruction quality. If a reliable upstream cloud mask is available, its source and version should be fixed across all baselines.

C. Reconstruction architecture

A reconstruction encoder receives the cloudy image, mask, and optional auxiliary data. Multiscale blocks encode local textures and larger land patterns; the decoder predicts a candidate clear image ■. A conditional GAN discriminator may distinguish reconstructed patches from reference patches. A diffusion branch can be introduced after a stable baseline has been reproduced. When auxiliary SAR is available, a separate encoder and cross-attention fusion block can condition reconstruction without treating radar intensity as optical reflectance. The model should be band-aware and preserve the original LISS-IV channel meaning.

D. Clear-pixel preservation and output

The final product uses mask-aware blending: Y = (1 − M■) ■ X + M■ ■ ■, where ■ is element-wise multiplication. In practice, the mask should include cloud shadows and uncertain boundaries may be blended softly. This reduces unnecessary changes to measured clear pixels. For thick clouds, uncertainty should be high because many ground states may be compatible with the same cloudy input. A confidence layer should accompany the output and be preserved with geospatial metadata in the exported GeoTIFF. Original cloudy input and reconstructed output should both remain available for audit.

IV. TRAINING OBJECTIVE

Let X* be the clear reference, X the cloudy input, M the cloud mask, and Y the reconstructed output. A suitable objective combines pixel fidelity, structure, spectral consistency, adversarial realism, and optional temporal or SAR consistency. The cloud-region pixel loss is Lpix = ||M ■ (Y − X*)||1. It focuses learning on missing pixels rather than rewarding the model for copying already-clear areas. A clear-region preservation term is Lclear = ||(1 − M) ■ (Y − X)||1. Structural loss can use Lssim = 1 − SSIM(Y, X*).

Spectral loss can use spectral angle mapper (SAM) between reconstructed and reference pixel spectra. For spectra y and x*, SAM is the angle between the two spectral vectors; lower values indicate closer spectral direction. If a GAN is used, an adversarial term Ladv supports realistic texture but should receive a conservative weight. When temporal references exist, Ltemp can penalize implausible disagreement with aligned observations while accounting for genuine land-cover change. SAR consistency should compare structural features, not force radar backscatter to equal optical reflectance.

The combined objective is L = λpix Lpix + λclear Lclear + λssim Lssim + λsam Lsam + λadv Ladv + λaux Laux. Weights λ are selected on a validation set and reported with each experiment. Laux can include temporal consistency, SAR-guided structure, or a boundary penalty. All losses must be calculated on consistent reflectance scales; otherwise the optimizer may prioritize scale differences instead of meaningful reconstruction. Ablation studies should remove one loss or auxiliary input at a time to identify which components actually help.

Training should use early stopping based on validation metrics, checkpoint selection fixed before test evaluation, and a documented random seed. Data augmentation may include flips, rotations, moderate brightness changes, and cloud-mask variation, provided transformations remain physically reasonable for the sensor. If synthetic cloud masks are used, vary their shape, opacity, and shadow displacement. Do not mix test-scene imagery into training through adjacent tiles or repeated acquisitions.

V. EXPERIMENTAL DESIGN AND METRICS

A fair experiment should compare the following baselines: B1 nearest-neighbour or bilinear filling; B2 multitemporal compositing; B3 U-Net with L1 loss; B4 U-Net with structural loss; B5 U-Net plus adversarial loss; B6 SAR-optical fusion when aligned SAR exists; and B7 conditional diffusion if compute permits. All models must use the same scene-level train/validation/test split and the same cloudy masks. Results should be reported separately for thin cloud, thick cloud, shadow, vegetation, built-up land, water, and changed surfaces.

Peak signal-to-noise ratio (PSNR) measures pixel error on a defined dynamic range. Structural similarity (SSIM) compares local structure. Root mean square error (RMSE) measures absolute reconstruction error. SAM captures spectral-vector angular deviation and is important for multispectral use. Report metrics over cloud-covered pixels and over the full image; full-image scores can appear deceptively good when most pixels are already clear. Include mask metrics, inference time, parameter count, memory use, and downstream land-cover classification or segmentation consistency.

Visual inspection should include easy and difficult cases, edge transitions, thin and thick cloud, and scenes with ground change. For stochastic models, evaluate multiple samples or provide uncertainty estimates. If a model is used for flood or crop monitoring, compare downstream outputs against reference labels to ensure reconstructed imagery does not degrade the task. Statistical summaries should include mean and standard deviation across test scenes, not only the best example.

No performance numbers are reported here because a verified LISS-IV dataset, baseline runs, and model outputs were not supplied. The next step is to run the protocol and replace this statement with measured results, confidence intervals, and figures. Values from other sensors or publications must not be presented as results for this project.

VI. COMPARATIVE STUDY

Method family	Strengths	Limitations	Recommended role
Interpolation	Fast; easy to reproduce	Blur and weak large-gap recovery	Baseline only
Temporal filling	Uses actual observations	Stale references / ground change	Use if near-date data exist
U-Net / CNN	Efficient spatial reconstruction	Needs representative data; may blur	Primary baseline
Conditional GAN	Can generate sharp texture	Hallucination and instability	Optional comparison
SAR fusion	All-weather structural cues	Alignment and modality gap	Auxiliary branch if SAR exists
Attention / transformer	Long-range and cross-band context	Memory and compute cost	Fusion enhancement
Diffusion	Flexible conditional generation	Slower inference; plausible errors	Advanced comparison
Tensor factorization	Uses temporal/spectral structure	Limited semantic modelling	Non-neural comparator

The literature does not establish one universally best method. Temporal methods are attractive when the ground has not changed; SAR contributes structure but not true optical colour; GAN and diffusion models can fill complex gaps but risk inventing content. For LISS-IV, the defensible approach is a staged comparison with explicit sensor adaptation and clear-pixel preservation. Diffusion should be justified by measured gains over a compact U-Net/GAN baseline, not chosen only because it is generative AI.

VII. LIMITATIONS, ETHICS, AND REPRODUCIBILITY

A reconstructed image is an estimate, not a recovered measurement. This distinction matters in flood mapping, crop assessment, and land-use change detection: a model could erase a small water body, invent a road, or smooth a field boundary. The system should mark reconstructed pixels, export uncertainty, retain the original input, and label outputs as reconstructed. Training data should cover geographic and seasonal diversity, and evaluation should report failure cases, not only attractive examples.

For reproducibility, publish the dataset manifest, sensor/band mapping, preprocessing parameters, train/validation/test scene IDs, random seeds, model configuration, loss weights, checkpoint-selection rule, evaluation script, and software versions. Record whether images are synthetic-cloud pairs or real cloudy scenes. Licensing and access conditions for satellite data must be checked before redistribution. Use the same output grid, projection, radiometric scale, and no-data policy for every baseline.

VIII. CONCLUSION AND FUTURE WORK

This paper presented an IEEE-style research framework and literature synthesis for generative-AI-based cloud removal and reconstruction in LISS-IV imagery. The comparative review indicates that robust solutions should combine cloud-aware masking, spatial reconstruction, temporal or SAR guidance when available, preservation of clear pixels, and spectral evaluation. A compact U-Net should be the first reproducible baseline; conditional GAN and diffusion variants can then be compared under the same split and metrics. Future work includes LISS-IV-specific fine-tuning, uncertainty calibration, multisensor co-registration, edge-aware spectral losses, and validation on downstream geospatial tasks. The central requirement is scientific trustworthiness: a visually plausible image is not necessarily a spectrally correct image.

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APPENDIX A — IMPLEMENTATION AND REPRODUCIBILITY PLAN

A.1 Suggested end-to-end workflow

Step 1: inventory imagery. For each scene, record sensor, acquisition date, geographic footprint, band order, pixel size, radiometric scale, coordinate reference system, cloud percentage, and access conditions. Keep the original files read-only and generate checksums so that preprocessing changes can be traced. Step 2: align imagery. Reproject auxiliary scenes to the target grid, document resampling methods, and verify alignment visually at roads, rivers, field boundaries, and other stable features. Step 3: prepare patches. Generate overlapping tiles and exclude patches dominated by no-data pixels. Step 4: create or validate masks. Keep cloud, cloud shadow, and invalid-data masks separate where possible. Step 5: split by scene/date before patch generation to avoid train/test leakage.

A.2 Recommended experiment registry

Every run should have a unique experiment ID and a machine-readable configuration. Record the model family, input bands, auxiliary data, patch dimensions, batch size, optimizer, learning rate, number of epochs, loss weights, random seed, training hardware, software versions, checkpoint rule, and test split. Store predictions and metric outputs under the same run ID. A model should not be selected by test-set performance; use the validation set and then evaluate the chosen checkpoint once on the held-out test scenes.

A.3 Suggested output package

For each input scene, save: (i) original cloudy GeoTIFF; (ii) cloud/shadow mask; (iii) reconstructed multispectral GeoTIFF; (iv) confidence or uncertainty raster; (v) visual preview with identical stretch for input and output; and (vi) a JSON/CSV manifest with model version and metrics. Preserve geotransform, projection, pixel dimensions, band names, no-data value, and reflectance scaling. Do not export a display-stretched PNG as the sole scientific output.

A.4 Quality-control checks

Before accepting a prediction, verify that clear pixels remain unchanged within a documented tolerance, reconstructed values stay within plausible sensor ranges, no-data regions remain no-data, band order is preserved, output geometry matches the input, and tile seams are not visible. Flag scenes with large cloud fractions or uncertain outputs. Compare spectral signatures for representative vegetation, water, soil, and built-up regions. Record known failure modes rather than silently smoothing them away.

APPENDIX B — RISKS, ABLATIONS, AND REPORTING CHECKLIST

B.1 Risk register

Risk	Why it matters	Mitigation / evidence
Hallucinated structures	Generated roads or rivers can mislead downstream maps	Mask-aware blending; confidence raster; human review
Spectral drift	Colour may look right but band values can be wrong	SAM and band-wise RMSE; reflectance-scale checks
Temporal mismatch	Older images may not reflect floods, construction, or crop growth	Date-aware reference selection; report temporal gap
SAR modality gap	Radar backscatter is not optical reflectance	Separate encoders; cross-attention; structure-aware losses
Data leakage	Nearby patches inflate test metrics	Split by scene/date before patching
Synthetic-to-real gap	Synthetic clouds may not model real atmospheric effects	Hold out real cloudy scenes for qualitative review
Overclaiming performance	No LISS-IV test results yet	Report only measured results; label planned work clearly

B.2 Essential ablation studies

Run the same model with (1) pixel loss only; (2) pixel plus SSIM; (3) pixel, SSIM, and SAM; (4) adversarial loss added; (5) temporal input removed; and (6) SAR input removed when applicable. These tests identify whether each component adds measurable value. Report metric differences and inference cost. For stochastic diffusion, report the number of sampling steps and whether results represent a single sample or an average across samples.

B.3 Final submission checklist

- ■ Replace all five author placeholders and institutional details.
- ■ Verify bibliographic metadata and replace discovery links [22]–[24] with exact records or remove them.
- ■ Confirm that every reference is cited in the body and all in-text citations resolve to the numbered list.
- ■ Add dataset details, real model diagrams, measured results, and representative output figures.
- ■ Use the official template of the target IEEE conference or journal before submission.
- ■ State the exact contribution of each team member and acknowledge data providers appropriately.
- ■ Do not describe this literature framework as completed experimental validation.